Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/30/2007
PI Name:	Britt, Anne Bagg Ph.D.		
Project Title:	Arabidopsis As A Model For The Study Of DNA Double Strand Break (DSB) Repair And DSB Induction By Space Radiation		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	95616-8535	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NCE to 12/31/2006, per A. Chu 7/2006.		
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Task Description:	Ionizing radiation induces oxidative damage, which affects all cellular components. However, the most significant form of damage induced by IR, in terms of cell death, mutagenesis, carcinogenesis, and effects on the cell cycle, is clustered DNA damage, often resulting in the formation of DNA double strand breaks. The DNA damage induced by high mass, high energy (HZE) space radiation is though to be more complex than that induced by gamma radiation, as HZE particles generate a denser trail of ionization events in their wake. We proposed to determine whether the same DNA repair pathways required for resistance to gamma radiation are also required for resistance to HZE particles. Because of the importance of double strand break (DSB) repair in prevention of carcinogenic “loss of heterozygosity” events, we have focused on DSB repair pathways. However, we are also interested in signal transduction pathways involved in the detection of IR induced damage- these pathways are important initiators of responses to IR, including cell cycle arrest, induction of repair, and induction of cell death. Thus we are investigating the effects of mutations in the nonhomologous end joining genes DNA ligase IV (LigIV), and Ku80, the endonuclease XPF, and the damage response genes ATM, ATR, and SOG. All of these genes are known to affect gamma resistance in mammals, fungi, and plants. Because knockouts in these genes are lethal in mammals (and, in some cases, in fungi), we are using the plant Arabidopsis as a model system. We are using the NSRL facility at Brookhaven to compare the sensitivity of these lines to gamma vs. 1GeV Fe particles. We plan to determine which DNA double strand break genes, previously identified as involved in resistance to gamma radiation, are also required for resistance to HZE radiation. The HZE component of cosmic radiation is recalcitrant to shielding and may result in a significant increase in risk of cancer in individuals undergoing long-term (months to years) exposure. The identification of genes involved in resistance to cosmic radiation may enable us to determine which individuals are especially susceptible to the carcinogenic effects of HZE and gamma radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identification of genes involved in resistance to various types of ionizing radiation has direct applications to the identification of cancer-susceptible individuals and the determination of appropriate treatment regimens for particular tumors,
Task Progress:	Effects of gamma vs HZE on plant growth: We irradiated Arabidopsis seeds of various genotypes and wild-type backgrounds in doses ranging from 25 to 200 Gy. Approximately 40 seeds were then sown on agar plates and the plates digitally imaged 8 days later. We used the application NIH Image to trace the roots and enter the length into Excel for statistical analyses. In no case did irradiation prohibit germination- at these doses Arabidopsis embryos do not die. Germination is driven by cell expansion, not cell division, but elongation of the root to lengths greater than approx. 1.5 mm requires cell division. At the highest doses of Fe ions (200 Gy), cell expansion as well as cell division was inhibited. Approximately twice this dose of gamma would be required to induce this effect on cell elongation. In order to contrast sensitivity in a quantitative fashion, we interpolated our dose-response curves to estimate the dose required to inhibit root growth at 8 days by 50%. As shown in Table I below, most mutants that are sensitive to gamma radiation are also sensitive to HZE. In almost all cases, the biological effectiveness of HZE was about 3-fold that of gamma radiation, suggesting the all of the mutations tested play similarly important roles in resistance to the effects of both types of radiation on root growth. The RBE numbers, however, need additional statistical analysis to determine their significance. The checkpoint genes ATR and ATM differed widely in their effects on resistance to HZE. Plants defective in ATR were relatively resistant to HZE vs. gamma (RBE = 2.7, rather than 3.7), while ATM knockouts were relatively sensitive (RBE = 4.5). Correct interpretation of this result would depend on whether growth inhibition in these mutants is due to the prolonged imposition of a checkpoint, or to the failure of a checkpoint and resulting cell death. The LOH data presented below in Figure 2 suggests that the relatively HZE-resistant growth of atr vs. atm may be due to the loss of the ATR-imposed checkpoint, resulting in enhanced loss of heterozygosity in the prematurely dividing cells. We also find that the sog mutation, a defect in a transcription factor required for checkpoint response to gamma radiation in the apical meristem, has little effect on root meristem response to either gamma or HZE.
Bibliography Type:	Description: (Last Updated: 12/29/2020)
Articles in Peer-reviewed Journals	Huefner ND, Yoshiyama K, Friesner JD, Conklin PA, Britt AB. "Genomic stability in response to high versus low linear energy transfer radiation in Arabidopsis thaliana." Front Plant Sci. 2014 May 20;5:206. eCollection 2014. http://dx.doi.org/10.3389/fpls.2014.00206 ; PubMed PMID: 24904606; PubMed Central PMCID: PMC4033213 , May-2014